

Our residuals are this:

$$S = \sum_{p \neq q} \sum_{mn} d_{pq}^{mn} - g_p^{ma} M^{ab} \overline{g_q^{bn}}. \quad (1)$$

p and q are baselines, m and n are correlations ($\in r, l$). Repeated indices are summed. What we minimize is the norm of S , that is, $S\overline{S}$, or $|S|^2$.

$$\frac{\partial |S|^2}{\partial m^{rr}} = \frac{\partial |S|^2}{\partial m^{rl}} = \frac{\partial |S|^2}{\partial m^{lr}} = \frac{\partial |S|^2}{\partial m^{ll}} = 0$$

Our sum of normed residuals *should* not depend on the model. If this is not the case, then, we are overfitting. Our solved gains are not universal and will only work with the model.

So when we set each of the above derivatives to zero, we get a system of four equations in four unknowns. All unknowns are the model parameters. So we can solve for them and see if we recover back our model that we used to solve for the gain terms.

All summations $\sum$ over all the baselines.

$$\begin{aligned} m^{rr} & \sum g_p^{lr} g_q^{rl} \overline{g_p^{lr} g_q^{rl}} + g_p^{lr} g_q^{rr} \overline{g_p^{lr} g_q^{rr}} + g_p^{rr} g_q^{rl} \overline{g_p^{rr} g_q^{rl}} + g_p^{rr} g_q^{rr} \overline{g_p^{rr} g_q^{rr}} \\ m^{rl} & \sum g_p^{lr} g_q^{rl} \overline{g_p^{lr} g_q^{ll}} + g_p^{lr} g_q^{rr} \overline{g_p^{lr} g_q^{ll}} + g_p^{rr} g_q^{rl} \overline{g_p^{rr} g_q^{ll}} + g_p^{rr} g_q^{rr} \overline{g_p^{rr} g_q^{ll}} \\ m^{lr} & \sum g_p^{ll} g_q^{rl} \overline{g_p^{lr} g_q^{rl}} + g_p^{ll} g_q^{rr} \overline{g_p^{lr} g_q^{rr}} + g_p^{rl} g_q^{rl} \overline{g_p^{rr} g_q^{rl}} + g_p^{rl} g_q^{rr} \overline{g_p^{rr} g_q^{rr}} \\ m^{ll} & \sum g_p^{ll} g_q^{rl} \overline{g_p^{lr} g_q^{ll}} + g_p^{ll} g_q^{rr} \overline{g_p^{lr} g_q^{ll}} + g_p^{rl} g_q^{rl} \overline{g_p^{rr} g_q^{ll}} + g_p^{rl} g_q^{rr} \overline{g_p^{rr} g_q^{ll}} \\ & = \sum d_{pq}^{rr} g_q^{rr} \overline{g_p^{rr}} + d_{pq}^{rl} g_q^{rl} \overline{g_p^{rr}} + d_{pq}^{lr} g_q^{rr} \overline{g_p^{lr}} + d_{pq}^{ll} g_q^{rl} \overline{g_p^{lr}} \end{aligned} \quad (2)$$

$$\begin{aligned} m^{rr} & \sum g_p^{lr} g_q^{ll} \overline{g_p^{lr} g_q^{rl}} + g_p^{lr} g_q^{rr} \overline{g_p^{lr} g_q^{rr}} + g_p^{rr} g_q^{ll} \overline{g_p^{rr} g_q^{rl}} + g_p^{rr} g_q^{lr} \overline{g_p^{rr} g_q^{rr}} \\ m^{rl} & \sum g_p^{lr} g_q^{ll} \overline{g_p^{lr} g_q^{ll}} + g_p^{lr} g_q^{rr} \overline{g_p^{lr} g_q^{ll}} + g_p^{rr} g_q^{ll} \overline{g_p^{rr} g_q^{ll}} + g_p^{rr} g_q^{lr} \overline{g_p^{rr} g_q^{ll}} \\ m^{lr} & \sum g_p^{ll} g_q^{ll} \overline{g_p^{lr} g_q^{rl}} + g_p^{ll} g_q^{rr} \overline{g_p^{lr} g_q^{rr}} + g_p^{rl} g_q^{ll} \overline{g_p^{rr} g_q^{rl}} + g_p^{rl} g_q^{lr} \overline{g_p^{rr} g_q^{rr}} \\ m^{ll} & \sum g_p^{ll} g_q^{ll} \overline{g_p^{lr} g_q^{ll}} + g_p^{ll} g_q^{rr} \overline{g_p^{lr} g_q^{ll}} + g_p^{rl} g_q^{ll} \overline{g_p^{rr} g_q^{ll}} + g_p^{rl} g_q^{lr} \overline{g_p^{rr} g_q^{ll}} \\ & = \sum d_{pq}^{rr} g_q^{lr} \overline{g_p^{rr}} + d_{pq}^{rl} g_q^{ll} \overline{g_p^{rr}} + d_{pq}^{lr} g_q^{lr} \overline{g_p^{lr}} + d_{pq}^{ll} g_q^{ll} \overline{g_p^{lr}} \end{aligned} \quad (3)$$

$$\begin{aligned} m^{rr} & \sum g_p^{lr} g_q^{rl} \overline{g_p^{ll} g_q^{rl}} + g_p^{lr} g_q^{rr} \overline{g_p^{ll} g_q^{rr}} + g_p^{rr} g_q^{rl} \overline{g_p^{rl} g_q^{rl}} + g_p^{rr} g_q^{rr} \overline{g_p^{rl} g_q^{rr}} \\ m^{rl} & \sum g_p^{lr} g_q^{rl} \overline{g_p^{ll} g_q^{ll}} + g_p^{lr} g_q^{rr} \overline{g_p^{ll} g_q^{ll}} + g_p^{rr} g_q^{rl} \overline{g_p^{rl} g_q^{ll}} + g_p^{rr} g_q^{rr} \overline{g_p^{rl} g_q^{ll}} \\ m^{lr} & \sum g_p^{ll} g_q^{rl} \overline{g_p^{ll} g_q^{rl}} + g_p^{ll} g_q^{rr} \overline{g_p^{ll} g_q^{rr}} + g_p^{rl} g_q^{rl} \overline{g_p^{rl} g_q^{rl}} + g_p^{rl} g_q^{rr} \overline{g_p^{rl} g_q^{rr}} \\ m^{ll} & \sum g_p^{ll} g_q^{rl} \overline{g_p^{ll} g_q^{ll}} + g_p^{ll} g_q^{rr} \overline{g_p^{ll} g_q^{ll}} + g_p^{rl} g_q^{rl} \overline{g_p^{rl} g_q^{ll}} + g_p^{rl} g_q^{rr} \overline{g_p^{rl} g_q^{ll}} \\ & = \sum d_{pq}^{rr} g_q^{rr} \overline{g_p^{rl}} + d_{pq}^{rl} g_q^{rl} \overline{g_p^{rl}} + d_{pq}^{lr} g_q^{rr} \overline{g_p^{ll}} + d_{pq}^{ll} g_q^{rl} \overline{g_p^{ll}} \end{aligned} \quad (4)$$

$$\begin{aligned} m^{rr} & \sum g_p^{lr} g_q^{ll} \overline{g_p^{lr} g_q^{rl}} + g_p^{lr} g_q^{rr} \overline{g_p^{ll} g_q^{rr}} + g_p^{rr} g_q^{ll} \overline{g_p^{rl} g_q^{rl}} + g_p^{rr} g_q^{lr} \overline{g_p^{rr} g_q^{rr}} \\ m^{rl} & \sum g_p^{lr} g_q^{ll} \overline{g_p^{lr} g_q^{ll}} + g_p^{lr} g_q^{rr} \overline{g_p^{ll} g_q^{ll}} + g_p^{rr} g_q^{ll} \overline{g_p^{rl} g_q^{ll}} + g_p^{rr} g_q^{lr} \overline{g_p^{rl} g_q^{ll}} \\ m^{lr} & \sum g_p^{ll} g_q^{ll} \overline{g_p^{lr} g_q^{rl}} + g_p^{ll} g_q^{rr} \overline{g_p^{ll} g_q^{rr}} + g_p^{rl} g_q^{ll} \overline{g_p^{rl} g_q^{rl}} + g_p^{rl} g_q^{lr} \overline{g_p^{rr} g_q^{rr}} \\ m^{ll} & \sum g_p^{ll} g_q^{ll} \overline{g_p^{ll} g_q^{ll}} + g_p^{ll} g_q^{rr} \overline{g_p^{ll} g_q^{ll}} + g_p^{rl} g_q^{ll} \overline{g_p^{rl} g_q^{ll}} + g_p^{rl} g_q^{lr} \overline{g_p^{rl} g_q^{ll}} \\ & = \sum d_{pq}^{rr} g_q^{lr} \overline{g_p^{rl}} + d_{pq}^{rl} g_q^{ll} \overline{g_p^{rl}} + d_{pq}^{lr} g_q^{lr} \overline{g_p^{ll}} + d_{pq}^{ll} g_q^{ll} \overline{g_p^{ll}} \end{aligned} \quad (5)$$